

Shiyu Hu

Research Fellow, Nanyang Technological University, Singapore

 1995.10.29  shiyu.hu@ntu.edu.sg  hushiyu1995  huuuuusy
 <https://huuuuusy.github.io/>  Google Scholar  ResearchGate



Research Interests

I study the dynamic states of humans and AI in open environments. As AI moves from answering single questions to carrying out sustained tasks, from long-horizon agents to video and world models and AI working alongside people, the hard part is no longer a correct answer at one moment but keeping the right state: which target is being followed, how long a description holds, which evidence still counts, and where the person stands. Using human performance and human judgment as reference, I evaluate where this state holds and breaks, model systems that keep it, and apply them to UAV vision, video understanding, and education.

Across Conditions	Whether a capability still holds when task and environment conditions change: user-defined task spaces, condition-level benchmarks with human baselines, capability profiles that predict behavior in new settings, and tracking for nighttime UAV vision
Over Time	Whether a system that answers correctly has kept the right state: target identity after disappearance, descriptions that hold only for a stretch of video, language cues that go stale, and long and streaming video understanding
Between Humans and AI	What counts as correct when people disagree or work with AI: human judgments as reference distributions, human-machine comparison on the same tasks, LLM-based virtual students that behave like learners, and AI in education

Each direction runs from evaluation to modeling to application: evaluate where a state holds and breaks, model systems that keep, update, or leave it open, and test them in real settings.

Work Experience

- 2024.08–Present **Research Fellow, School of Physical and Mathematical Sciences (SPMS), Nanyang Technological University (NTU)**
- **Research:** Computer Vision, Multimodal AI, and AI for Education
 - **PI:** Assoc. Prof. Kang Hao Cheong (IEEE Senior Member)
- 2018.02–2018.12 **Research Assistant, University of Hong Kong (HKU)**
- **Research:** High Performance Computing, Heterogeneous Computing
 - **PI:** Prof. Choli Wang
- 2016.08–2016.09 **Research Intern, Institute of Electronics, Chinese Academy of Sciences (CASIE)**

Education

- 2019.09–2024.01 **Ph.D., Institute of Automation, Chinese Academy of Sciences (CASIA)**
- **Field:** Computer Application Technology
 - **Supervisor:** Prof. Kaiqi Huang (IAPR Fellow, IEEE Senior Member)
 - **Co-supervisor:** Prof. Xin Zhao (IEEE Senior Member)
 - **Thesis title:** *Research of Intelligence Evaluation Techniques for Single Object Tracking*
 - **Thesis committee:** Profs. Jianbin Jiao, Yuxin Peng (CAAI/CIE/CSIG Fellow), Yao Zhao (IEEE/IET Fellow), Yunhong Wang (IEEE/IAPR/CCF Fellow), and Ming Tang
 - **Thesis defense grade:** Excellent
- 2017.09–2019.07 **M.Sc., Department of Computer Science, University of Hong Kong (HKU)**
- **Field:** Computer Science
 - **Supervisor:** Prof. Choli Wang
 - **Thesis title:** *NightRunner: Deep Learning for Autonomous Driving Cars after Dark*
 - **Thesis defense grade:** A+

Education (continued)

- 2013.09–2017.07 **B.E., Elite Class in School of Information and Electronics, Beijing Institute of Technology (BIT)**
- **Field:** Information Engineering
 - **Supervisor:** Prof. Senlin Luo
 - **Thesis title:** *Text Sentiment Analysis Based on Deep Neural Network*
 - **Thesis defense grade:** Excellent
- 2015.07–2015.08 **Summer Session, University of California, Berkeley (UC Berkeley)**
- **Course:** New Media
 - **Course grade:** A

Research Projects and Platforms

Funded Research Projects

- 2025–Present **Singapore MOE Tertiary Education Research Fund** (MOE2024-TRF-004)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)
- 2023–2025 **Singapore MOE Tertiary Education Research Fund** (MOE2022-TRF-029)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)
- 2024–2025 **2023 Intelligent Education PhD Research Fund**
- **Research Contributor**, Shanghai Institute of AI Education, ECNU

Research Platforms

- 2019.11–Present **VideoCube / MGIT Platform** <http://videocube.aitestunion.com>
- **Role:** builder and maintainer; evaluation infrastructure for the TPAMI 2023 and NeurIPS 2023 studies
 - **Impact (July 2026):** 1.66M+ visits, 1.9k+ data users, 720+ registered trackers, 960+ result submissions
- 2021.07–Present **SOTVerse / VLTVerse Platform** <http://metaverse.aitestunion.com>
- **Role:** builder and maintainer; task-space evaluation platform for the IJCV 2024 study
 - **Impact (July 2026):** 267k+ visits
- 2022.05–Present **BioDrone Platform** <http://biodrone.aitestunion.com/>
- **Role:** builder and maintainer; benchmark platform for real-world drone tracking (IJCV 2024)
 - **Impact (July 2026):** 541k+ visits
- 2020.07–Present **GOT-10k Platform** <http://got-10k.aitestunion.com/>
- **Role:** maintainer; evaluation platform for the TPAMI 2021 study
 - **Impact (July 2026):** 7.56M+ visits, 11k+ data users, 32.7k+ registered trackers, 406k+ result submissions

Challenges

- 2023.05–2023.11 **Hislopvision Challenge** <http://hislopvision.aitestunion.com/>
- Organized the Hislopvision track of the 3rd High-speed and Low-power Visual Understanding Challenge at PRCV 2023; participating teams from Tsinghua University, Beijing Institute of Technology, and Jilin University
- 2021.01–2021.04 **Cell Tracking Challenge** <https://celltrackingchallenge.net/>
- 2nd place on Fluo-C2FL-MSc+ and 3rd place on Fluo-C2FL-Huh7 (rankings as of Oct. 2023)

Research Software

- 2018.02–2018.12 **Darknet-Cross** <https://github.com/huuuuusy/Darknet-Cross>
- Lightweight deep learning framework for heterogeneous computing: cross-platform acceleration for Android and Ubuntu across mobile and desktop GPUs; HKU master's thesis project

Academic Activities and Services

Tutorials

Organizer and speaker of all tutorials listed below.

- 2025.10.26 **28th European Conference on Artificial Intelligence (ECAI 2025, CCF-B)**
Title: From Benchmarking to Trustworthy AI: Rethinking Evaluation Methods Across Vision and Complex Systems
Location: Bologna, Italy
- 2025.10.05 **IEEE International Conference on Systems, Man, and Cybernetics (SMC 2025, CCF-C)**
Title: The Synergy of Large Language Models and Evolutionary Optimization on Complex Networks
Location: Vienna, Austria
- 2025.08.18 **34th International Joint Conference on Artificial Intelligence (IJCAI 2025, CCF-B)**
Title: Human-Centric and Multimodal Evaluation for Explainable AI: Moving Beyond Benchmarks
Location: Montreal, Canada
- 2024.12.09 **17th Asian Conference on Computer Vision (ACCV 2024, CCF-C)**
Title: From Machine-Machine Comparison to Human-Machine Comparison: Adapting Visual Turing Test in Visual Object Tracking
Location: Hanoi, Vietnam
- 2024.12.01 **27th International Conference on Pattern Recognition (ICPR 2024, CCF-C)**
Title: Visual Turing Test in Visual Object Tracking: A New Vision Intelligence Evaluation Technique based on Human-Machine Comparison
Location: Kolkata, India
- 2024.10.27 **31st IEEE International Conference on Image Processing (ICIP 2024, CCF-C)**
Title: An Evaluation Perspective in Visual Object Tracking: from Task Design to Benchmark Construction and Algorithm Analysis
Location: Abu Dhabi, United Arab Emirates

Talks and Mini-Symposia

- 2026.11 **Talk, College of Liberal Arts, Wenzhou-Kean University**
Series: CLA Invited Academic Talks
Title: Studying the Human Mind with AI: From Behavioral Analysis to Role Simulation
Location: Wenzhou, China
- 2026.10.04 **Talk, 14th International Conference on Image and Graphics (ICIG 2026)**
Forum: Visual Intelligence in Transition: From Physical Perception to Psychological Cognition
Title: From Object Tracking to Process Understanding: State Modeling in Dynamic Vision
Location: Singapore
- 2026.09.20 **Mini-Symposium, 5th International Nonlinear Dynamics Conference (NODYCON 2026)**
Title: Complex Network Systems and Large Language Models
Location: Sapienza University of Rome, Italy
- 2026.05.29 **Talk, Chinese Congress on Image and Graphics (CCIG 2026)**
Title: Visual Understanding Reliability in Open Environments: from Robust Perception to Semantic Consistency
Location: Guangzhou, China

Conference Organization and Program Committees

- Area Chair **15th International Conference on Learning Representations (ICLR, CCF-A)**
- TPC Member **34th Pacific Conference on Computer Graphics and Applications (PG 2026, CCF-B)**
- Publicity Chair **CSIG Annual Conference on Video and Image Security 2026**
Date: 2026.11.21
Location: Xiong'an, China

Academic Activities and Services (continued)

Editorial Roles

Lead Guest Editor	Electronics , Special Issues: Techniques and Applications of Multimodal Data Fusion – 1st and 2nd Editions
Associate Editor	Innovation and Emerging Technologies

Peer Review

Conferences	NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, ACL, AAAI, IJCAI, ACM MM, ICRA, and AISTATS
Journals	ACM Computing Surveys; IEEE Transactions on Image Processing; SCIENCE CHINA Information Sciences; Pattern Recognition; Transactions on Machine Learning Research; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Vehicular Technology; Information Fusion; Visual Intelligence; Engineering Applications of Artificial Intelligence; Expert Systems with Applications; Neurocomputing; Knowledge-Based Systems

Professional Memberships

Member	Committee: Technical Committee on Video and Image Security (CSIG-TCVIS) Societies: Institute of Electrical and Electronics Engineers (IEEE); Chinese Association for Artificial Intelligence (CAAI); and China Computer Federation (CCF)
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Student Mentoring and Co-Supervision

Joint publications with mentees include papers at CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, ACL, EMNLP, AAAI, IJCAI, and ICRA, and in IJCV, TCSVT, and Pattern Recognition; ongoing mentoring is listed first.

Ph.D. Students

2023.08–Present	Yiping Ma , East China Normal University
2023.08–Present	Dailing Zhang , Institute of Automation, Chinese Academy of Sciences
2024.08–Present	Xuchen Li , Institute of Automation, Chinese Academy of Sciences
2024.08–Present	Yipei Wang , Southeast University
2025.01–Present	Shukun Jia , Southeast University
2025.01–Present	Buyuan Zhu , Nanyang Technological University, Singapore
2025.08–Present	Zi Ye , University of Science and Technology Beijing
2025.08–Present	Yihan Meng , University of Science and Technology Beijing
2026.02–Present	Shunya Hirashima , Nanyang Technological University, Singapore
2022.08–2025.06	Meiqi Wu , University of Chinese Academy of Sciences
2023.04–2026.06	Xiaokun Feng , Institute of Automation, Chinese Academy of Sciences

M.S. Students

2024.07–Present	Xuzhao Li , Beijing Institute of Technology
2024.09–Present	Panxi Xu , University of Science and Technology Beijing
2025.07–Present	Hongfei Ma , University of Science and Technology Beijing
2022.05–2023.07	Yiping Ma , Nanjing Normal University
2022.08–2024.07	Yipei Wang , Southeast University
2024.07–2025.01	Yuqi Cui , University of Science and Technology Beijing
2025.02–2025.11	Nguyen Khanh Truong , Nanyang Technological University, Singapore

B.E. Students

2022.09–2023.08	Junyou Zhu , University of Chinese Academy of Sciences
2022.09–2023.08	Lihang Hu , University of Chinese Academy of Sciences
2022.09–2023.08	Dailing Zhang , Southeast University
2023.04–2024.07	Xuchen Li , Beijing University of Posts and Telecommunications

Student Mentoring and Co-Supervision (continued)

2024.09–2025.05	Le Ying Tan , Nanyang Technological University, Singapore
2024.09–2025.07	Zi Ye , University of Science and Technology Beijing
2024.10–2025.07	Jinlin Ma , Nanyang Technological University, Singapore
2024.10–2026.05	Yuxiao Li , Nanyang Technological University, Singapore
2024.12–2025.07	Yihan Meng , University of Science and Technology Beijing
2025.01–2025.05	Yihui Jiang , University of Science and Technology Beijing
2025.04–2025.07	Shunya Hirashima , Nanyang Technological University, Singapore
2025.07–2025.09	Anastasiia Bohachenko , Nanyang Technological University, Singapore (Global Connect Fellow)
2025.08–2026.05	Jia Ling Ow , Nanyang Technological University, Singapore
2025.08–2026.05	Jian Feng Ho , Nanyang Technological University, Singapore

Honors and Awards

Professional and Graduate Honors

2026	Reviewer Award , 43rd International Conference on Machine Learning (ICML 2026)
2025	IEEE SMCS TEAM Program Award , IEEE Systems, Man, and Cybernetics Society
2024	Best Paper Honorable Mention , 3rd Workshop on Vision Datasets Understanding and DataCV Challenge in CVPR 2024 Beijing Outstanding Graduate , Beijing Municipal Education Commission (Top 5%)
2023	China National Scholarship , Ministry of Education of the People's Republic of China (Top 1%) First Prize of Climbing Scholarship , Institute of Automation, Chinese Academy of Sciences
2022	Merit Student , University of Chinese Academy of Sciences

Undergraduate Honors

Competitions	National First Prize , Contemporary Undergraduate Mathematical Contest in Modeling (2015, Top 1%); First Prize , Mathematical Modeling Competition, Beijing Institute of Technology (2015)
Scholarships	Academic Scholarship , Beijing Institute of Technology (2013–2017); College Scholarship , Chinese Academy of Sciences (2016)
Student Honors	Excellent Innovative Student (2017), Outstanding Student Cadre (2015), and Outstanding Student (2014), Beijing Institute of Technology
Leadership & Service	Youth Day Competition, Beijing Institute of Technology: Excellent League Member (2016), Outstanding League Cadre (2015), Outstanding Youth League Branch (2015, Team Leader), and Top 10 Activities (2015, Team Leader); Summer Social Practice: Outstanding Individual (2015) and Second Prize (2015, Team Leader)

Skills & Interests

Programming	Python, MATLAB, C, Java, JavaScript, HTML/CSS, Shell, Git, and $\text{\LaTeX}$; LLM-assisted coding
Systems	Research-platform and Flask/SQLite web development; Linux, Android, and heterogeneous computing
Leadership	Research supervision; grant writing; project delivery; cross-institutional collaboration; academic writing, peer review, tutorials, and presentations
Languages	Chinese (native) and English
Hobbies	Art, writing, LEGO, fitness, and former amateur marathoner

Research Publications

Publications: 1 monograph; 42 journal and conference papers (14 in CCF-A venues; 9 as first or corresponding author: TPAMI, IJCV ×2, NeurIPS, ICCV Highlights, ICRA, ICMR, IET-CVI, and a JIG survey); 4 workshop papers; 11 preprints

Citations: Google Scholar, Oct. 2026: 1,703 citations, h-index 18, i10-index 27

Notation: * equal contribution; ✉ corresponding author; CCF-T1/T2/T3: CCF 2025 tiers for Chinese journals; other journals outside the CCF list: JCR quartile and 2025 IF

Monograph

- [B1] X. Zhao, **S. Hu**, and X. Yin, *Visual Object Tracking: An Evaluation Perspective*. Springer, 2025, ISBN: 978-981-96-4558-9. DOI: 10.1007/978-981-96-4558-9

First- and Corresponding-Author Publications

- [A1] **S. Hu**, X. Zhao, L. Huang, and K. Huang, “Global instance tracking: Locating target more like humans,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI, CCF-A)*, vol. 45, no. 1, pp. 576–592, 2023. DOI: 10.1109/TPAMI.2022.3153312
- [A2] **S. Hu**, X. Zhao, and K. Huang, “SOTVerse: A user-defined task space of single object tracking,” *International Journal of Computer Vision (IJCV, CCF-A)*, vol. 132, pp. 872–930, 2024. DOI: 10.1007/s11263-023-01908-5
- [A3] X. Zhao, **S. Hu**✉, Y. Wang, J. Zhang, Y. Hu, R. Liu, H. Ling, Y. Li, R. Li, K. Liu, and J. Li, “BioDrone: A bionic drone-based single object tracking benchmark for robust vision,” *International Journal of Computer Vision (IJCV, CCF-A)*, vol. 132, pp. 1659–1684, 2024. DOI: 10.1007/s11263-023-01937-0
- [A4] **S. Hu**, D. Zhang, M. Wu, X. Feng, X. Li, X. Zhao, and K. Huang, “A multi-modal global instance tracking benchmark (MGIT): Better locating target in complex spatio-temporal and causal relationship,” *Advances in Neural Information Processing Systems (NeurIPS, CCF-A)*, vol. 36, pp. 25 007–25 030, 2023.
- [A5] X. Feng*, **S. Hu***, X. Li, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, “ATCTrack: Aligning target-context cues with dynamic target states for robust vision-language tracking,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV, CCF-A; Highlights)*, 2025, pp. 19 850–19 861.
- [A6] X. Li*, X. Li*, and **S. Hu**✉, “MATrack: Efficient multiscale adaptive tracker for real-time nighttime UAV operations,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA, CCF-B)*, 2026.
- [A7] X. Li*, X. Li*, and **S. Hu**✉, “DARTer: Dynamic adaptive representation tracker for nighttime UAV tracking,” in *Proceedings of the International Conference on Multimedia Retrieval (ICMR, CCF-B)*, 2025.
- [A8] **S. Hu**, X. Zhao, and K. Huang, “Visual intelligence evaluation techniques for single object tracking: A survey,” *Journal of Image and Graphics (《中国图象图形学报》) (CCF-T2)*, vol. 29, no. 8, pp. 2269–2302, 2024. DOI: 10.11834/jig.230498
- [A9] Y. Wang, J. Zhang, Y. Wang, **S. Hu**✉, B. Shen, Z. Hou, and W. Zhou, “Improved SAR aircraft detection algorithm based on visual state space models,” *IET Computer Vision (IET-CVI, CCF-C)*, vol. 19, no. 1, e70032, 2025.

Collaborative Publications

- [C10] P. Zhao, **S. Hu**, J. Zhang, and M. Li, “META-PAP: Meta-learning for prompt-aware preference pairing in LLM alignment,” *Advances in Neural Information Processing Systems (NeurIPS, CCF-A)*, 2026.
- [C11] X. Li, X. Li, **S. Hu**, and K. Huang, “Select less, reason more: Prioritizing evidence purity for video reasoning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, CCF-A)*, 2026.
- [C12] X. Li, X. Li, **S. Hu**, K. Huang, and W. Zhang, “CausalStep: A benchmark for explicit stepwise causal reasoning in videos,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, CCF-A; Oral)*, 2026.

- [C13] X. Li, X. Li, **S. Hu**, Y. Guo, and W. Zhang, "VerifyBench: A systematic benchmark for evaluating reasoning verifiers across domains," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, CCF-A; Oral)*, 2026.
- [C14] X. Feng, H. Yu, M. Wu, **S. Hu**, J. Chen, C. Zhu, J. Wu, X. Chu, and K. Huang, "NarrLV: Towards a comprehensive narrative-centric evaluation for long video generation models," in *Proceedings of the International Conference on Learning Representations (ICLR, CCF-A)*, 2026.
- [C15] X. Li, X. Li, J. Gao, R. Pi, **S. Hu**, and W. Zhang, "Look less, reason more: Rollout-guided adaptive pixel-space reasoning," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL, CCF-A)*, 2026.
- [C16] S. Jia, **S. Hu**, Y. Cao, F. Yang, X. Lu, and X. Lu, "Tracking by detection and query: An efficient end-to-end framework for multi-object tracking," *Pattern Recognition (PR, CCF-B)*, 2026.
- [C17] M. Wu, Y. Zhao, X. Li, **S. Hu**, Y. Cai, J. Wu, W. Wang, and K. Huang, "Talk with your fingers: A depth-aware benchmark for air-writing recognition," *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT, CCF-B)*, 2026. DOI: 10.1109/TCSVT.2026.3670256
- [C18] S. Jia, **S. Hu**, Y. Wang, X. Cheng, Y. Cao, and X. Lu, "COAL: Counterfactual and observation-enhanced alignment learning for discriminative referring multi-object tracking," in *the 35th International Joint Conference on Artificial Intelligence (IJCAI, CCF-B)*, 2026.
- [C19] D. Zhang, **S. Hu**, H. Fu, X. Feng, Y. Wang, K. H. Cheong, and K. Huang, "Rethinking temporal modeling in visual object tracking via decoupled auxiliary supervision," in *Proceedings of the European Conference on Computer Vision (ECCV, CCF-B)*, 2026.
- [C20] M. Peng, H. Gu, Y. Ma, **S. Hu**, and X. Huang, "Fake news court: A multi-agent adversarial framework for robust detection of LLM-generated fake news," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP, CCF-B)*, 2026.
- [C21] X. Li, X. Li, R. Pi, **S. Hu**, J. Zhao, and J. Gao, "Beyond accuracy: Evaluating grounded visual evidence in thinking with images," in *Findings of the Association for Computational Linguistics: EMNLP 2026 (EMNLP Findings)*, 2026.
- [C22] K. Li, B. Parikh, H. Yue, **S. Hu**, S. W. Tan, W. Y. Low, X. Su, and K. H. Cheong, "Global-local semi-supervised modeling for retinal layer boundary estimation in OCT," in *the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC, CAAI-B)*, 2026.
- [C23] X. Li, X. Li, **S. Hu**, Z. Zhang, and K. H. Cheong, "Constraint-driven evolution of multimodal video intelligence: A network and system perspective," *IEEE Transactions on Network Science and Engineering (TNSE, JCR Q1, IF 7.3)*, 2026.
- [C24] S. Hirashima, **S. Hu**, and K. H. Cheong, "Beyond hyper-rationality: Persona-driven heterogeneity in networked agent-based systems," *IEEE Transactions on Network Science and Engineering (TNSE, JCR Q1, IF 7.3)*, 2026.
- [C25] Y. Ma, M. Peng, **S. Hu**, B. Zhu, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "From observation to generation: A complex-systems perspective on generative human-AI social systems," *Chaos, Solitons & Fractals (CSF, JCR Q1, IF 5.7)*, 2026.
- [C26] L. Tan, B. Zhu, **S. Hu**, A. Mishra, D. J. Yeo, and K. H. Cheong, "CalcTutor: Multi-agent LLM grading of handwritten mathematics with RAG-grounded feedback for adaptive learning support," *Mathematics (JCR Q1, IF 2.3)*, 2026.
- [C27] Center for AI Safety, Scale AI, and HLE Contributors Consortium (including **Shiyu Hu**), "A benchmark of expert-level academic questions to assess AI capabilities," *Nature (JCR Q1, IF 56.1)*, vol. 649, pp. 1139–1146, 2026. DOI: 10.1038/s41586-025-09962-4

- [C28] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, “CSTrack: Enhancing RGB-X tracking via compact spatiotemporal features,” in *Proceedings of the International Conference on Machine Learning (ICML, CCF-A)*, 2025.
- [C29] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, “Enhancing vision-language tracking by effectively converting textual cues into visual cues,” in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP, CCF-B)*, 2025.
- [C30] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, “Artificial intelligence-enabled adaptive learning platforms: A review,” *Computers & Education: Artificial Intelligence (C&E:AI, JCR Q1, IF 23.4)* (No. 3 of 210 in *Artificial Intelligence; highest JIF among AI-in-education journals*), p. 100 429, 2025.
- [C31] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, “A comprehensive review on automated grading systems in STEM using AI techniques,” *Mathematics (JCR Q1, IF 2.3)*, vol. 13, no. 17, 2025.
- [C32] Y. Ma, X. Li, **S. Hu**, S. Liu, and K. H. Cheong, “Trustworthy AI in education: Framework, cases, and governance strategies,” *Innovation and Emerging Technologies (JCR Q2, IF 2.3)*, vol. 12, p. 2 550 026, 2025.
- [C33] K. Huang, Y. Kang, C. Yan, **S. Hu**, L. Wang, T. Tao, and W. Gao, “A review of intelligent psychological assessment based on interactive environment,” *Chinese Mental Health Journal (《中国心理卫生杂志》)* (CSCD Core, PKU Core), 2025.
- [C34] D. Zhang, **S. Hu**, X. Feng, X. Li, M. Wu, J. Zhang, and K. Huang, “Beyond accuracy: Tracking more like human via visual search,” *Advances in Neural Information Processing Systems (NeurIPS, CCF-A)*, vol. 37, pp. 2629–2662, 2024.
- [C35] X. Feng, X. Li, **S. Hu**, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, “MemVLT: Visual-language tracking with adaptive memory-based prompts,” *Advances in Neural Information Processing Systems (NeurIPS, CCF-A)*, vol. 37, pp. 14 903–14 933, 2024.
- [C36] J. Zhang, T. Zhao, **S. Hu**, and X. Zhao, “Robust single-particle Cryo-EM image denoising and restoration,” in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP, CCF-B)*, 2024.
- [C37] M. Wu, K. Huang, Y. Cai, **S. Hu**, Y. Zhao, and W. Wang, “Finger in camera speaks everything: Unconstrained air-writing for real-world,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT, CCF-B)*, vol. 34, no. 9, pp. 8602–8613, 2024.
- [C38] M. Wu, Y. Kang, X. Li, **S. Hu**, X. Chen, Y. Kang, W. Wang, and K. Huang, “VS-LLM: Visual-semantic depression assessment based on LLM for drawing projection test,” in *Chinese Conference on Pattern Recognition and Computer Vision (PRCV, CCF-C)*, 2024, pp. 232–246.
- [C39] X. Feng, **S. Hu**, X. Chen, and K. Huang, “A hierarchical theme recognition model for sandplay therapy,” in *Chinese Conference on Pattern Recognition and Computer Vision (PRCV, CCF-C)*, 2023, pp. 241–252.
- [C40] Y. Wang, **S. Hu**, and X. Zhao, “Rethinking similar object interference in single object tracking,” in *International Conference on Computer Science and Artificial Intelligence (CSAI, EI; Oral)*, 2023, pp. 251–258.
- [C41] Y. Zhang, C. Liu, W. Chen, X. Xu, F. Wang, H. Li, **S. Hu**, and X. Zhao, “Revisiting instance search: A new benchmark using cycle self-training,” *Neurocomputing (Neu, CCF-C)*, vol. 501, pp. 270–284, 2022. DOI: 10.1016/j.neucom.2022.06.027
- [C42] K. Huang, X. Zhao, Q. Li, and **S. Hu**, “Visual turing: The next development of computer vision in the view of human-computer gaming,” *Journal of Graphics (《图学报》)* (CCF-T3), vol. 42, no. 3, p. 339, 2021. DOI: 10.11996/JG.j.2095-302X.2021030339

Workshop Papers

- [W1] Y. Ma*, **S. Hu***, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, “Learning to be taught: A structured SOEI framework for modeling and evaluating personality-aligned virtual student agents,” in *AAAI Conference on Artificial Intelligence AI4Edu Workshop (AAAIW, CCF-A Workshop; Oral)*, 2026.

- [W2] Y. Ma*, **S. Hu***, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, “Redefining educational simulation: EduVerse as a user-defined and developmental multi-agent simulation space,” in *AAAI Conference on Artificial Intelligence AI4Edu Workshop (AAAIW, CCF-A Workshop; Oral)*, 2026.
- [W3] B. Zhu*, **S. Hu***, Y. Ma, Y. Zhang, and K. H. Cheong, “From objective to subjective: A benchmark for virtual student abilities,” in *AAAI Conference on Artificial Intelligence AI4Edu Workshop (AAAIW, CCF-A Workshop; Oral)*, 2026.
- [W4] X. Li, X. Feng, **S. Hu**, M. Wu, D. Zhang, J. Zhang, and K. Huang, “DTLLM-VLT: Diverse text generation for visual language tracking based on LLM,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition 3rd VDU Workshop (CVPRW, CCF-A Workshop; Oral, Best Paper Honorable Mention)*, 2024.

Preprints

- [P1] X. Li*, **S. Hu***, X. Feng, J. Zhao, and K. Huang, “How to respond, how to memorize, how to be fast: A survey of streaming video understanding,” *SSRN preprint*, 2026.  URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7008298
- [P2] **S. Hu***, X. Li*, X. Li, J. Zhang, Y. Wang, X. Zhao, and K. H. Cheong, “FIOVA: A multi-annotator benchmark for human-aligned video captioning,” *arXiv preprint arXiv:2410.15270*, 2024.
- [P3] Y. Ma*, **S. Hu***, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, “When LLMs learn to be students: The SOEI framework for modeling and evaluating virtual student agents in educational interaction,” *arXiv preprint arXiv:2410.15701*, 2024.
- [P4] Y. Ma*, **S. Hu***, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, “EduVerse: A user-defined multi-agent simulation space for education scenario,” *arXiv preprint arXiv:2510.05650*, 2025.
- [P5] B. Zhu*, **S. Hu***, Y. Ma, Y. Zhang, and K. H. Cheong, “EduPersona: Benchmarking subjective ability boundaries of virtual student agents,” *arXiv preprint arXiv:2510.04648*, 2025.
- [P6] Y. Wang*, **S. Hu***, S. Jia, P. Xu, H. Ma, Y. Ma, J. Zhang, X. Lu, and X. Zhao, “SOI is the root of all evil: Quantifying and breaking similar object interference in single object tracking,” *arXiv preprint arXiv:2508.09524*, 2025.
- [P7] X. Li*, **S. Hu***, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, “How texts help? a fine-grained evaluation to reveal the role of language in vision-language tracking,” *arXiv preprint arXiv:2411.15600*, 2024.
- [P8] X. Li, X. Li, J. Zhao, and **S. Hu**✉, “STEMVerse: A dual-axis diagnostic framework for STEM reasoning in large language models,” *arXiv preprint arXiv:2602.02497*, 2026.
- [P9] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, “DTVLT: A multi-modal diverse text benchmark for visual language tracking based on LLM,” *arXiv preprint arXiv:2410.02492*, 2024.
- [P10] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, “Visual language tracking with multi-modal interaction: A robust benchmark,” *arXiv preprint arXiv:2409.08887*, 2024.
- [P11] **S. Hu**, X. Zhao, Y. Wang, Y. Shan, and K. Huang, *Nearing or surpassing: Overall evaluation of human-machine dynamic vision ability*, 2023.  URL: https://openreview.net/forum?id=LGbzYw_pnsc

References

Prof. Kaiqi Huang

Professor, IAPR Fellow, IEEE Senior Member, 10,000 Talents Program - Leading Talents
Director of Center for Research on Intelligent Systems and Engineering (CRISE)
National Laboratory of Pattern Recognition (NLPR)
Institute of Automation, Chinese Academy of Sciences (CASIA)
✉ kqhuang@nlpr.ia.ac.cn

Prof. Choli Wang

Honorary Professor
Department of Computer Science, University of Hong Kong (HKU)
✉ clwang@cs.hku.hk
✉ choliwang@gmail.com

Prof. Xin Zhao

Professor, IEEE Senior Member, Beijing Science Fund for Distinguished Young Scholars
School of Computer and Communication Engineering, University of Science and Technology Beijing (USTB)
✉ xinzhao@ustb.edu.cn
✉ xzhaopersonal@foxmail.com

Prof. Kang Hao Cheong

Associate Professor, IEEE Senior Member
Assistant Dean (Continuing Education and Training), College of Science
Nanyang Technological University (NTU)
✉ kanghao.cheong@ntu.edu.sg